The Sudbury dikes of the Grenville Front region: paleomagnetism, petrochemistry, and K–Ar age studies

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Received December 2, 1976

Revision accepted for publication March 8, 1977

Northwest trending diabase dikes having alkali olivine basalt chemistry occur abundantly in the Southern Structural Province of the Canadian Shield of Ontario. In several subregions these dikes can be traced across the Grenville Front for a short distance into the Grenville Province. Within the Southern Province the dikes possess a westerly directed primary magnetization yielding a paleomagnetic pole of high precision and accuracy (168° W, 2½° S, $A_{95} = 2½°$). The geological age of these dikes is probably about 1250 Ma. At distances ranging from 2–8 km (northwest) of the Grenville Front these dikes are remagnetized and are characterized by anomalously high apparent K–Ar ages. The acquisition of this later magnetization having a direction of 111°, +26½° with a corresponding pole at 16½° W, 4° S, $A_{95} = 6½°$ takes place without any general mineralogical expression of metamorphism within the dikes. A few dikes sharing the above direction of magnetization were intruded during this time of reheating at about 1000 Ma. A short dike was located in the Grenville Front region southwest of Sudbury which possesses tholeiitic chemistry, early Paleozoic apparent K–Ar ages, and a cleaned remanence direction of 125°, +59° with a corresponding pole at 42° W, 10° N. These early Paleozoic dikes may be more extensive within the Grenville Province.

Des dykes de diabase de direction nord-ouest dont la chimie est celle d'un basalte à olivine alcalin se rencontrent en abondance dans la province structurale du Sud dans le bouclier canadien en Ontario. Dans plusieurs sous-régions, ces dykes peuvent être retracés à travers le front de Grenville sur une courte distance dans la province de Grenville. Dans la province du Sud, ces dykes possèdent une magnétisation primaire dirigée vers l'ouest avec un pôle paléomagnétique localisé avec grande précision et exactitude (168° O, 2½° S, $A_{95} = 2½°$). L'âge géologique de ces dykes est probablement d'environ 1250 Ma. A des distances allant de 2 à 8 km (au nord-ouest) du front de Grenville, ces dykes sont remagnétisés et sont caractérisés par des âges apparents K–Ar anormalement élevés. L'acquisition de cette dernière magnétisation a une direction de 111°, +26½° avec un pôle correspondant à 16½° O, 4° S, $A_{95} = 6½°$, et se produit sans aucune manifestation minéralogique de métamorphisme dans les dykes. Quelques dykes ayant cette direction de magnétisation one été mis en place durant cette période de réchauffement il y a environ 1000 Ma. Un dyke court situé dans la région du front de Grenville au sud-ouest de Sudbury possède une chimie tholéiitique, des âges apparents K–Ar datant du début du Paléozoïque et a une direction de rémanence nette de 125°, +59° avec un pôle correspondant à 42° O, 10° N. Ces dykes qui datent du début du Paléozoïque peuvent être plus extensifs dans la province de Grenville.

[Traduit par le journal]

Can. J. Earth Sci. **14**, 1867–1887 (1977)

Introduction

A number of previous paleomagnetic studies (Sopher 1963; Fahrig *et al.* 1965; Larochelle 1967) have shown that west-northwest trending olivine diabase dikes north and west of Sudbury share a common paleomagnetic direction of remanence (westerly with no or little dip component). Furthermore, from arguments based on the dispersion of the directional data (Larochelle 1967) and from baked contact studies (Sopher 1963; Schwarz 1977) the fossil remanence is likely to be a primary remanence acquired during original cooling of the dikes. Potassium–argon analyses of biotites from two of these dikes suggest an age of 1285 Ma[1]. (Fahrig *et al.* 1965). More recently Gates and Hurley (1973) have obtained a Rb–Sr isochron of 1460 ± 130 Ma on whole rock samples which had previously been used by Larochelle in his paleomagnetic study. Van Schmus (1965) obtained a Rb–Sr age of 1225 ± 25 Ma from an olivine diabase well to the west of the Sudbury area. This dike was sampled in the present study (sites 59 and 60) and it also yields a typical Sud-

[1]Ma = m.y. in SI units.

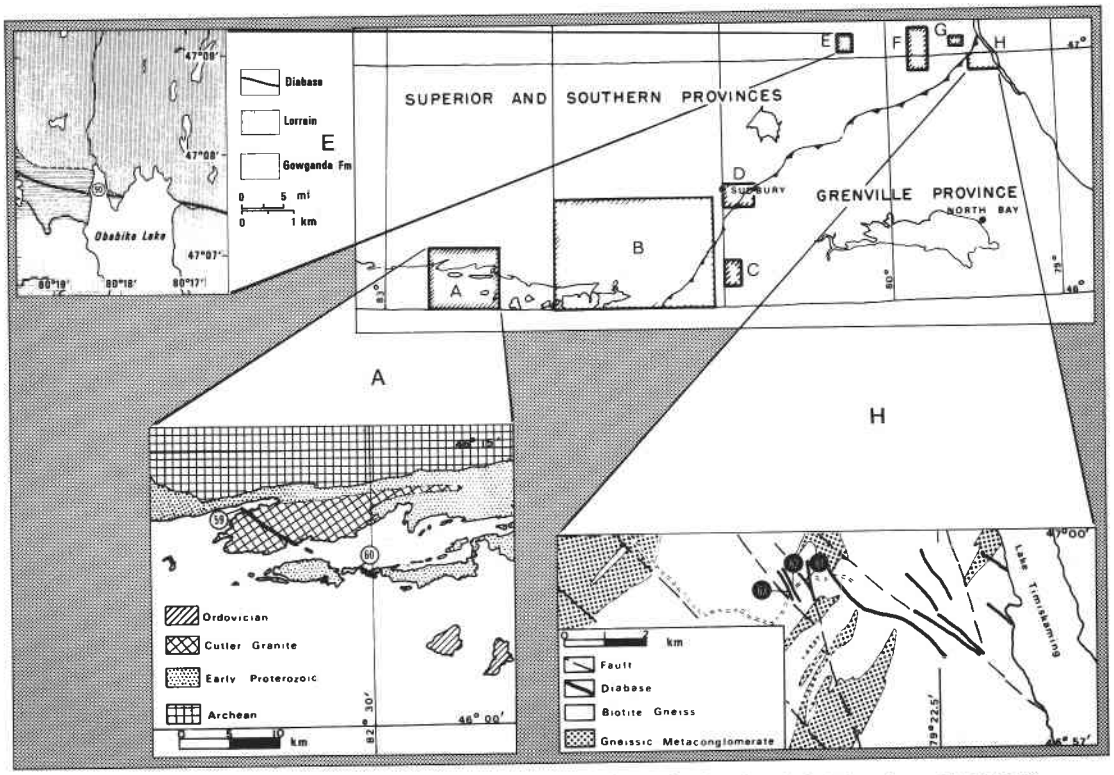


FIG. 1. Index map showing sampling regions. The geology of subregion A is taken from MAP 2108, Ontario Department of Mines, and from Robertson (1965). The geology of subregion E is generalized from Map 1954-1 (Township of Delhi), Ontario Department of Mines. The geology of subregion H is generalized from Lumbers (1971). Paleomagnetic sampling sites having westerly declinations and shallow inclinations, 'Sudbury dike direction', are enclosed by open circles. Paleomagnetic sampling sites having ESE declinations and downward inclinations of remanence, 'Grenville Front direction', are enclosed by solid circles.

bury dike paleomagnetic direction. Thus, from these studies, the age of the magnetization in these dikes falls in the time span 1460–1225 Ma.

Most of the above mentioned paleomagnetic and geochronologic studies were completed before systematic geological mapping had been completed in the region. Frarey and Cannon (1969) published the first modern map of a substantial subregion of region B of Fig. 1 and subsequently Card (1971*a*,*b*) published a compilation map of all of subregion B of Fig. 1, an outline of which is here reproduced as Fig. 2. These maps reveal that some of the olivine dikes of the Sudbury swarm continue to the south and east of the Huronian outcrop belt and into the Grenville Province. Since the olivine dikes in the Sudbury region to the north had previously been shown to retain a remanence older than the last metamorphism in the Grenville Province, the new geological maps suggested that answers to some yet unresolved paleomagnetic problems might be obtained from a paleomagnetic and petrologic study of the dikes in this region. Specifically, as the dikes are traced to the south and east and presumably become metamorphosed as the Grenville Front is approached and crossed, what happens to the original remanence in the dikes? Is it destroyed completely? Is it replaced by a new remanence acquired during the later metamorphism? Does part of the original remanence survive and is some new magnetization added so that one sees only a resultant of the two? At what grade or facies of metamorphism do these processes take place? How would this remagnetization correlate in a spatial sense with the lowering of apparent K–Ar ages?

Murthy (1971), in a study of generally east–west trending diabase dikes exposed to the east of subregion B of Fig. 1, described them as being of the quartz diabase type and found that stably

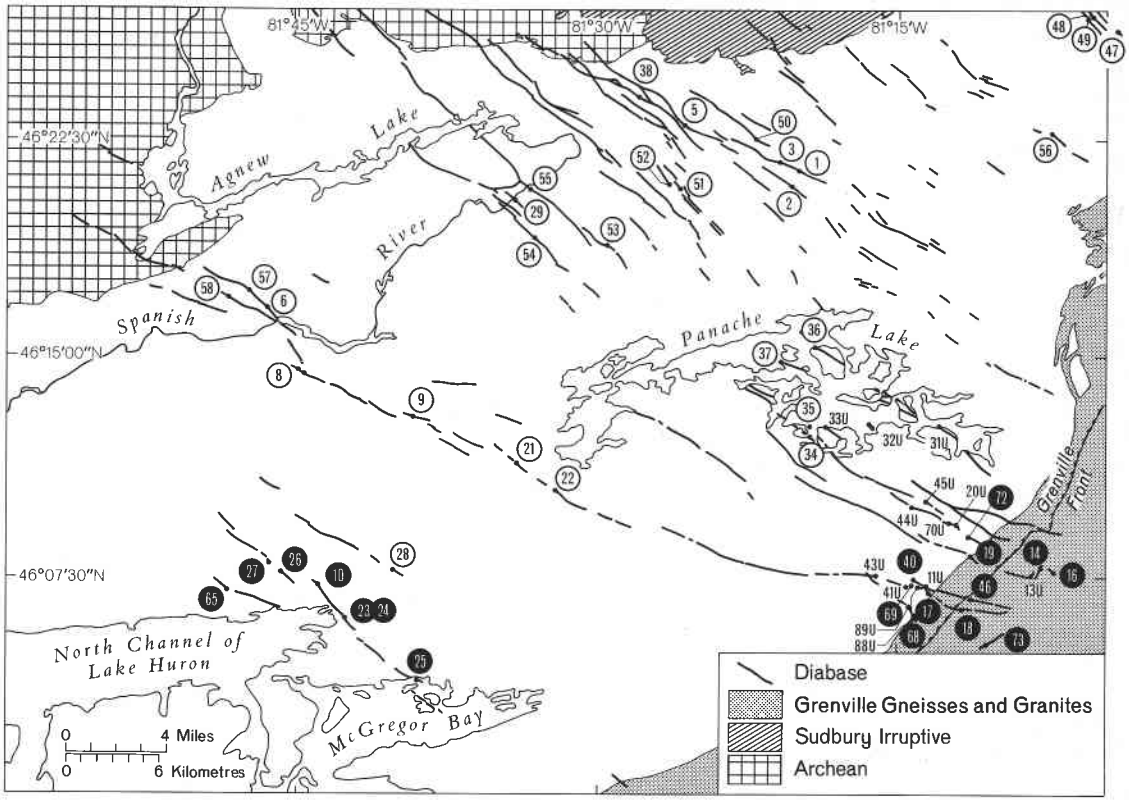


FIG. 2. An enlarged map of subregion B of Fig. 1 with the distribution of dikes and regional geology generalized from Card (1971*a*, *b*). Open and solid symbols enclosing sampling site numbers have the same meaning as in Fig. 1. Sites not readily classified as belonging to either of these two paleomagnetic groupings are followed by the letter 'U' for unclassified sites.

magnetized sites had a remanence direction of 110°, +45°, which is very different from that reported from the Sudbury dikes. Furthermore, Lumbers (1975, pp. 108–110) has presented geological evidence that several ages of diabase dikes may be present in the Southern Province – Grenville Province terrains of the Burwash map-area.

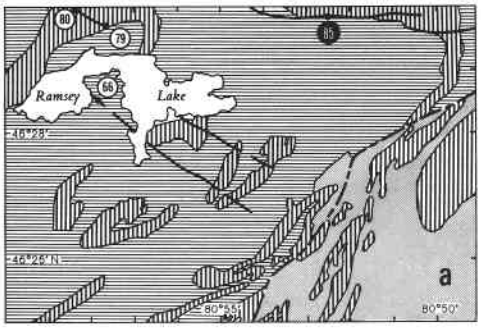
Accordingly, whole rock petrochemical studies were also undertaken to supplement the paleomagnetic and K–Ar data. The paleomagnetic information will be presented first, followed by the petrochemical and K–Ar data.

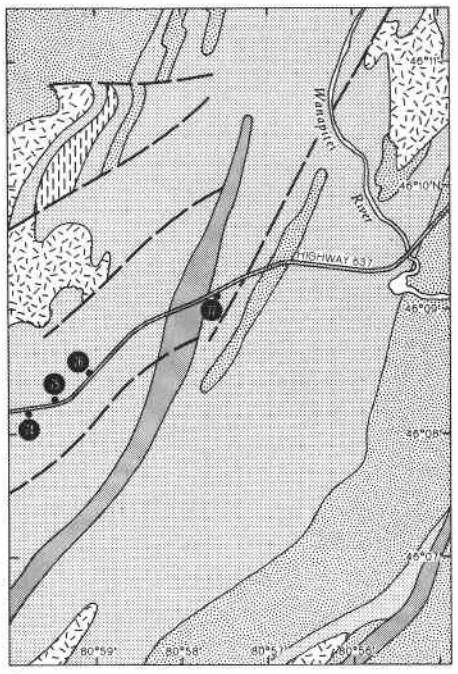
Paleomagnetic Field and Laboratory Methods

The initial sampling was concentrated in the area outlined in Fig. 2. However, as the spatial pattern of remanence directions in relation to the Grenville Front emerged it seemed worthwhile to sample in additional areas on either side of the Grenville Front to test the general applicability of the tentative conclusions reached from the data of the Panache Lake area.

Samples collected at sites shown on Figs. 1–5, were drilled *in situ* by means of a portable drill and oriented with a device similar to that described by Helsley (1967). Although magnetic bearings of the azimuthal component of orientation were recorded for all samples, sun sightings were obtained for all samples collected at 62 of the total of 79 sites within the diabases. For those 17 sites where sunsights were unavailable, magnetic bearings taken close to and at a distance from the outcrop or sightings to distant landmarks ensured that the orientations of these cores are accurate to within 3°. The plunge of all cores was recorded to the nearest degree.

Typically four cores (samples) were obtained at each site and generally each sample was long enough to provide two paleomagnetic specimens arbitrarily labelled 1 and 2. A polished thin section was prepared from one core from each site.





Specimen 1 of each sample from sites 1–46 and 61–63 was measured on a PAR SM-1 spinner magnetometer and subjected to AF demagnetization in 50 Oe (3.98×10^3 A/m)[2] steps to a maximum of 400 Oe (31.8×10^3 A/m) (peak). The apparatus used for the demagnetization of these specimens is similar in design to that described by Roy *et al.* (1973). The direction and intensity of remanence of the second specimen from each sample from sites 1–46 and 61–63 were measured on parastatic magnetometers and demagnetized in 50 Oe (3.98×10^3 A/m) steps to a routine maximum of 400 Oe (31.8×10^3 A/m) (peak). Selected second specimens were further demagnetized in 100 Oe (7.96×10^3 A/m) steps to a maximum of 1400 Oe (111.4×10^3 A/m) (peak). Most of the specimens from sites 47–93 were measured on a Schonstedt DSM-1 magnetometer. The earlier paleomagnetic work on the Sudbury dikes (e.g. Larochelle 1967) indicated that these rocks are characterized by low to moderate coercivity of remanence spectra and low optimum cleaning fields based on end-point and within-site directional precision criteria. Typically rocks of this type display increasing within-site and between-site disperison of the mean direction with increasing applied alternating field strengths above some optimum field. Much of this dispersive effect may be an intrinsic rock property but some may well be due to undesirable characteristics of the alternating field demagnetizing apparatus (Irving 1964). One possible type of dispersive effect may be a 'rotational remanent magnetization' (RRM) induced by unidirectional tumbling. Comparative data were obtained from the second specimens by demagnetization on the apparatus described by Wilson and Lomax (1972) which is designed to eliminate RRM.

Median destructive fields of natural remanent magnetization (NRM) were picked from demagnetization curves plotted for each specimen. Low field magnetic susceptibility measurements were made with a Bison Model 3101A susceptibility

[2]1 Oe = 79.6 A/m in SI units.

FIG. 3. Enlarged maps of subregions D and C of Fig. 1. The geology depicted on this figure is generalized from portions of Map 2271, Ontario Division of Mines, by S. B. Lumbers (1975). Sample site symbols have the same significance as in Fig. 1.

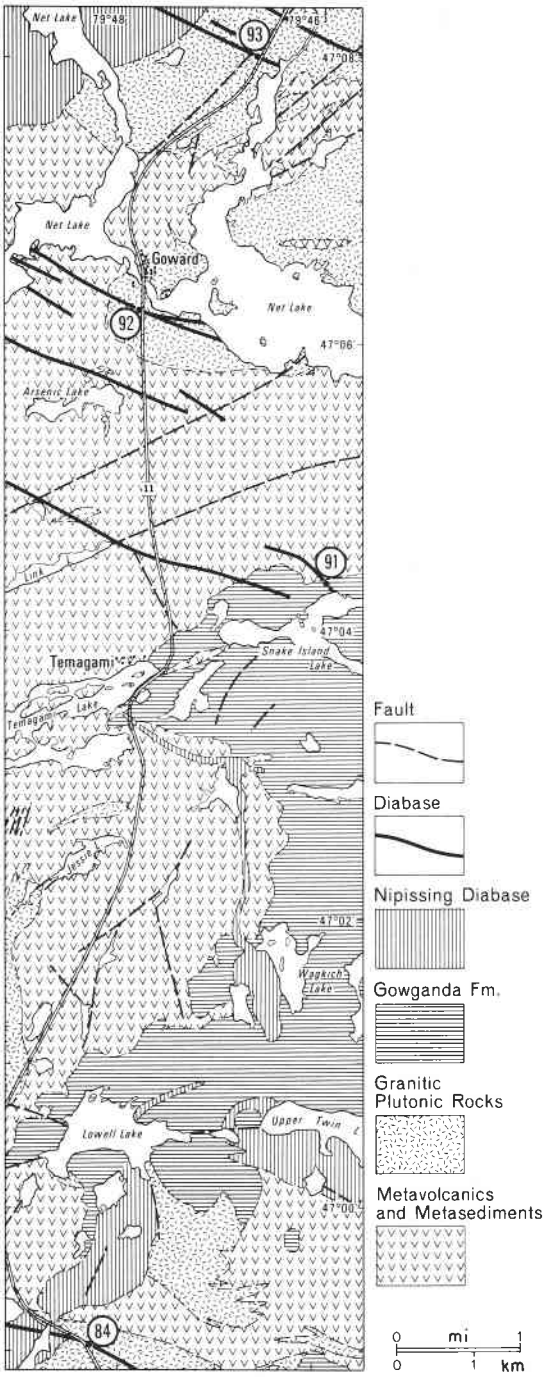


FIG. 4. Geological map of subregion F of Fig. 1 showing the location of sampling sites: 84, 91, 92, and 93. The geology depicted on this figure is generalized from Bennett (1969) and Bennett and Innes (1970).

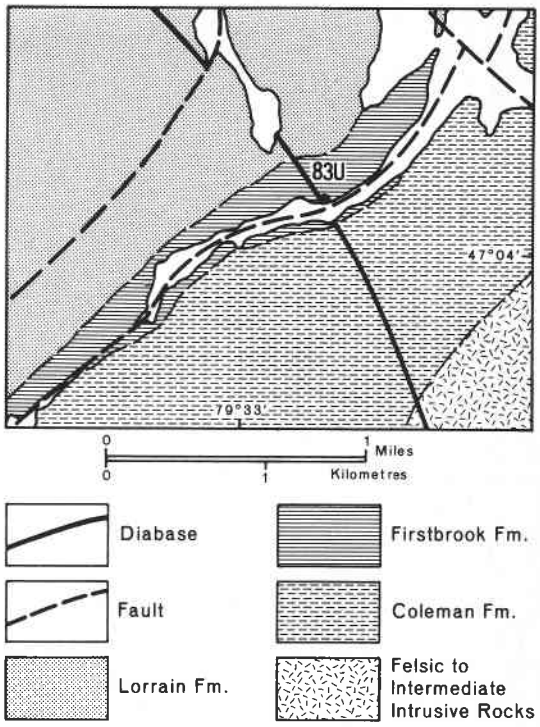


FIG. 5. Geological map of subregion G of Fig. 1. The geology depicted on this figure is taken from Ontario Department of Mines, Map 2194, by G. W. McIlwaine (1970). Site 83 is an 'unclassified' site in the preliminary paleomagnetic analysis; the detailed paleomagnetic behaviour of samples from site 83 is recorded on Fig. 8.

bridge and these results were used in calculating Koenigsberger ratios. Arithmic site-means and standard deviations were calculated from these data and these results together with high field Curie points for selected samples are recorded in Table 2.

Paleomagnetic Results

Within-site mean directions of remanence were calculated for all sites after each stage of demagnetization. For sites 1–46 and 61–63 this was done using three data sets: specimen 1 results only, specimen 2 results only, and with specimen 1 and specimen 2 first combined to give core mean (sample mean) as the basic datum. These latter results are presented in Table 1. Minimum dispersion within the range of a defined stable end point was used as the criterion for selection of the cleaned direction to represent each site. Comparison of specimen 1 and specimen 2 data revealed no systematic difference in

TABLE 1. Site-mean directional data

	NRM					After AF treatment							
Site	N	D	I	R	AF(Oe)	N	D	I	R	k	α_{95}	Pole position	
1	4	286.	−19.	2.780	150	4	274.	−02.	3.994	499.	4°	175° W	2° N
2	4	279.	00.	3.916	200	4	271.	−15.	3.994	529.	4°	177° W	5° S
3	4	292.	+43.	3.979	200	4	274.	+07.	3.996	843.	3°	172° W	5° N
5	2	274.	+34.	1.911	150	2	270.	−04.	1.987	76.	29°	173° W	1° S
6	4	266.	+37.	3.809	300	4	272.	+07.	3.977	129.	8°	171° W	4° N
3L	4	275.	+42.	—	150	4	272.	00.	—	57.	—	—	—
8	4	265.	+24.	3.991	300	4	261.	+08.	3.999	3487.	2°	163° W	3° S
9	4	272.	+29.	3.141	300	4	263.	00.	3.998	1582.	2°	167° W	5° S
10	4	112.	+36.	3.641	200	4	096.	+27.	3.965	85.	10°	6° W	6° N
11	5	122.	+79.	4.964	—	No stable end point							
13	4	321.	+52.	3.552	300	4	336.	+53.	3.252	4.	53°	—	—
14	2	172.	+31.	0.923	300	2	133.	+14.	1.976	42.	40°	29° W	22° S
16	4	068.	+32.	3.938	100	4	097.	+14.	3.963	82.	10°	1° W	0° N
17	4	227.	+39.	3.903	200	4	122.	+51.	3.839	19.	22°	35° W	4° N
18	4	069.	+71.	3.137	150	4	097.	+07.	3.702	10.	31°	1° E	2° S
19	4	255.	+64.	3.970	300	4	145.	+30.	3.941	51.	13°	45° W	20° S
20	4	178.	+80.	3.948		No stable end point							
21	4	284.	+28.	3.937	200	4	277.	−04.	3.961	77.	11°	178° W	3° N
22	4	264.	+14.	3.984	200	4	258.	−11.	3.997	924.	3°	167° W	12° S
23	4	097.	+41.	3.958	300	4	105.	+18.	3.977	133.	8°	9° W	4° S
24	4	097.	+19.	3.997	250	4	097.	+16.	3.999	6914.	1°	2° W	1° N
25	4	109.	+54.	3.954	200	4	111.	+28.	3.954	65.	12°	17° W	3° S
26	4	096.	+38.	3.997	250	4	103.	+22.	3.999	7316.	1°	9° W	1° S
27	4	108.	+58.	3.974	300	4	105.	+30.	3.987	240.	6°	14° W	2° N
28	4	241.	+35.	3.717	400	4	243.	+16.	3.967	90.	10°	146° W	12° S
29	4	270.	+43.	3.754	200	4	264.	+03.	3.967	92.	10°	166° W	3° S
31*	4	270.	+50.	3.622	350	4	254.	+62.	3.848	20.	21°	—	—
32*	4	300.	−53.	3.942	400	4	239.	−08.	3.664	9.	33°	—	—
33*	4	311.	−45.	3.995	300	4	305.	−40.	3.989	282.	6°	—	—
34	4	278.	−62.	3.802	250	4	239.	−10.	3.880	25.	19°	151° W	25° S
35	4	307.	−41.	3.992	400	4	264.	−08.	3.953	64.	12°	170° W	7° S
36	4	262.	−04.	1.169	300	4	251.	−23.	3.834	18.	22°	166° W	22° S
37	4	224.	+21.	3.967	150	4	253.	+08.	3.994	510.	4°	156° W	9° S
38	4	247.	+08.	3.079	300, 900	4	248.	00.	3.875	24.	19°	155° W	15° S
40	4	161.	+60.	3.874	300	4	122.	+47.	3.978	135.	8°	33° W	1° N
41	4	117.	+78.	3.958	—	No stable end point							
43	4	Random				Random							
44*	4	002.	+67.	3.686	300	4	355.	+66.	3.303	4.	50°	—	—
45*	4	225.	−01.	1.379	400	4	256.	−36.	3.475	6.	42°	—	—
46	4	069.	+79.	3.993	400	4	116.	+69.	3.996	714.	3°	45° W	23° N

TABLE 1. (*Concluded*)

Site	NRM					After AF treatment							
	N	*D*	*I*	*R*	AF(Oe)	*N*	*D*	*I*	*R*	*k*	α_{95}	Pole position	
47	4	231.	+50.	3.594	150	4	256.	+30.	3.963	80.	10°	150° W	2° N
48*	4	029.	−31.	2.509	700	4				Random			
48r	4	352.	+29.	2.869	150	4	281.	+33.	3.780	14.	26°	166° W	20° N
49	4	313.	+16.	2.735	150	4	271.	+12.	3.757	12.	27°	168° W	5° N
50	4	279.	+32.	3.974	200	4	270.	−05.	3.992	371.	5°	173° W	2° S
51	3	267.	+24.	2.991	150	3	264.	−01.	2.998	881.	4°	167° W	5° S
52	4	262.	+18.	3.983	250	4	263.	+07.	3.988	255.	6°	164° W	2° S
53	4	262.	+03.	2.461	250	4	272.	−14.	3.960	76.	11°	178° W	4° S
54	4	264.	+16.	3.977	200	4	261.	−05.	3.991	350.	5°	167° W	8° S
9L	4	262.	+16.	—	150	5	261.	−01.	—	73.	—	—	—
55	4	271.	+56.	3.941	150	4	264.	+02.	3.964	83.	10°	167° W	3° S
7L	4	237.	+69.	—	150	2	266.	+04.	—	44.	—	—	—
56	4	249.	+44.	3.541	150	4	258.	00.	3.987	233.	6°	162° W	8° S
57	4	301.	+50.	3.020	250	4	260.	+22.	3.957	69.	11°	157° W	2° N
58	4	264.	+29.	3.880	200	4	261.	+06.	3.859	21.	20°	163° W	4° S
59	4	307.	+47.	2.839	150	4	271.	−04.	3.968	93.	10°	175° W	1° S
60	4	268.	+21.	3.929	400	4	260.	−05.	3.995	646.	4°	167° W	9° S
61	4	101.	+66.	3.932	400	4	112.	+33.	3.975	118.	9°	17° W	1° S
62	4	029.	+41.	3.915	300	4	140.	+36.	3.950	60.	12°	41° W	14° S
63	4	124.	+36.	3.975	350	4	130.	+25.	3.991	344.	5°	29° W	15° S
65	4	064.	+64.	3.944	250	4	091.	+31.	3.954	65.	12°	4° W	11° N
66	4	281.	+40.	3.920	200	4	270.	+11.	3.980	149.	8°	167° W	4° N
68	6	038.	+78.	5.948	300	6	130.	+58.	5.561	11.	21°	44° W	6° N
69	4	091.	+81.	3.924	400	4	127.	+50.	3.954	66.	11°	38° W	1° N
70a*	5	323.	+67.	4.739	300	5	314.	+32.	3.864	3.	48°	—	—
70b*	3	149.	+75.	2.913	600	3	142.	+60.	2.980	101.	12°	—	—
72	4	305.	−71.	3.921	250	4	120.	+03.	3.834	18.	22°	15° W	19° S
73	4	347.	−06.	2.241	500	4	119.	+32.	3.981	159.	7°	24° W	6° S
74	3	077.	+51.	2.957	400	3	111.	+19.	2.975	81.	14°	13° W	7° S
75	4	063.	+07.	2.819	400	4	107.	+23.	3.983	182.	7°	12° W	3° S
76	4	108.	+46.	1.853	500	4	124.	+27.	3.955	67.	11°	26° W	11° S
77	3	043.	−48.	1.517	400	3	094.	+23.	2.979	97.	13°	2° W	6° N
79	4	277.	−25.	3.681	200	4	268.	−06.	3.988	255.	6°	172° W	4° S
80	4	275.	+28.	3.948	300	4	269.	−04.	3.978	137.	8°	172° W	2° S
83	4	107.	+52.	3.992						No stable end point			
84	4	266.	+02.	3.967	400	4	264.	−10.	3.972	107.	9°	169° W	8° S
85	4	051.	+63.	3.943	350	4	088.	+35.	3.852	20.	21°	3° W	15° N
88*	4	220.	−10.	0.757	1000					Random			
89	3	071.	−53.	2.632	900	3	108.	−56.	2.462	4.	—	—	—
90	4	266.	+05.	2.981	300	4	270.	+10.	3.984	186.	7°	167° W	4° N
91	4	275.	+33.	3.981	300	4	272.	+29.	3.984	187.	7°	161° W	13° N
92	3	274.	+04.	2.996	200	3	276.	−07.	2.996	582.	5°	177° W	2° N
93	4	281.	+09.	3.869	300	4	278.	−06.	3.965	85.	10°	178° W	3° N

NOTES: Sites 3L, 9L, and 7L are sites of Larochelle (1967) which are from the same outcrops as sites 6, 54, and 55 of this study. Sites with * may have been affected by lightning. Other 'U' type sites of Fig. 1 and Fig. 2 for which poles are not calculated are discussed in the text. D = declination of remanence in degrees, clockwise from true north. I = inclination of remanence in degrees, north-seeking directions below horizontal taken as positive. R = vector resultant derived from unit vectors of N cores. k = precision estimate of the mean direction (Fisher 1953). α_{95} = radius of the circle of confidence about the mean direction in degrees (Fisher 1953).

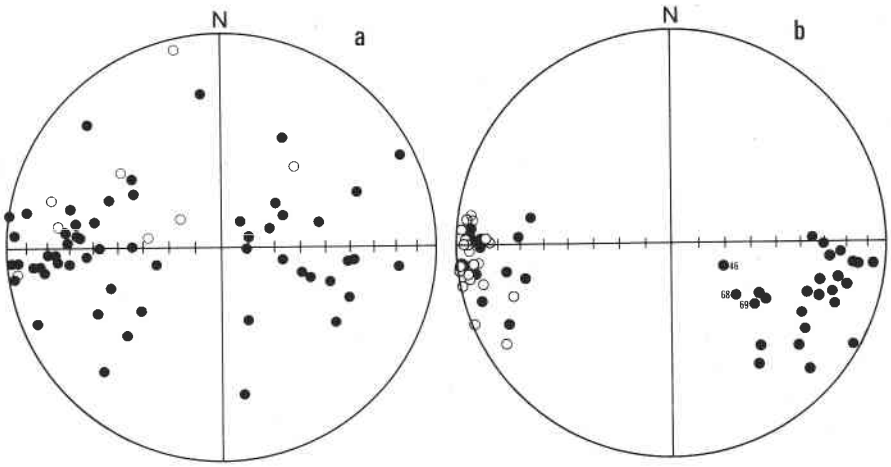


FIG. 6. Site-mean directional data for all classified sites of Figs. 1–5 before and after alternating field demagnetization at the fields indicated in Table 1. Solid (open) symbols represent north-seeking directions of remanence on the lower (upper) hemisphere on equal area projections. After cleaning (b) these sites form two groupings centered about mean directions of 265°, +02° and 113° + 30°. Sites 46, 68, and 69 are identified on this figure as they are petrographically and chemically distinct and have Early Paleozoic apparent ages.

optimum cleaning field, or site mean direction and associated statistics indicating that 'rotational remanent magnetizations' are not a cause of the dispersive effect seen in the majority of these diabases above the optimum cleaning field. For sites 47–60 and 65–93, specimen 1 served as a basic datum for the results recorded in Table 1.

Three of the sites of this study (6, 54, and 55) are common with three sites of Larochelle (1967) and his results for these sites (coded as 3L, 9L, and 7L) are included in Table 1. This interlaboratory comparison, which involves the complete process of paleomagnetic data gathering, provides an encouraging result.

Most of the sites yield fossil remanences which fall into one of two directional groups. These are graphically displayed in Fig. 6. The spatial distribution of these sites is coded in Figs. 1 through 5. No sites with the typical Sudbury dike remanence direction are observed within the Grenville Province. Sites possessing remanences with east-southeast declinations and downward inclinations are restricted to the Grenville Province and the adjacent part of the Southern Province. Furthermore, as can be seen on Fig. 6, the ESE-down magnetized group of sites forms a more dispersed group than do those sites which possess the 'Sudbury dike direction'. Evidence is given below that the ESE-down grouping includes: (1) remagnetized Sudbury alkali olivine diabase dikes, (2) alkali olivine diabase dikes intruded during this time of remagnetization (ca. 1.0 Ga [3]), and (3) minor volumes of olivine tholeiite diabase intruded in early Paleozoic time.

A number of sites cannot be classified as belonging to either of these two directional groups and they are coded as 'U' type sites on Fig. 2 and Fig. 5. Eight of these sites (31, 32, 33, 44, 45, 48, 70, and 88) possess aberrant directions of remanence although for some of them, the within-site precision is acceptable and they exhibit stable end points against AF fields up to 1000 Oe (79.6×10^3 A/m). All of these sites are characterized by very high initial intensities of remanence approaching the saturation remanence, high Q ratios, and low coercivities of remanence (Table 2). Large discrepancies between the sun azimuth and magnetic azimuth were recorded at these sites in the process of field collection. These sites are believed to have been remagnetized by lightning strikes and no paleomagnetic information can be extracted from them. In the course of the field work, site 48 was revisited and a second location (48r) was located where the local magnetic field was much less disturbed. These sample possess less aberrant magnetic properties (Table 2) and are included as providing useful paleofield information (Table 1). Two other sites, 36 and 38, appear to have

[3] 1 Ga = 1 b.y. in SI units.

been partially remagnetized by lightning but AF demagnetization removes most, if not all, of this component and they are accepted as providing paleofield information. Sites 13, 43, and 89 are rejected on the criterion of low within-site precision throughout all stages of demagnetization.

Spiralling Paleomagnetic Sites

Sites 11, 20, 41 (Fig. 2) and site 83 (Fig. 5) cannot be simply classified as belonging to either of the two main directional groupings but they provide significant information bearing on the age relationships of the two directional groups. They were classified as 'U' type sites in the preliminary analysis and are coded as such on Fig. 2 and Fig. 5 because no clearly defined end point of high precision was achieved. Their behaviour during demagnetization, illustrated in Figs. 7 and 8, is similar in that they first move away from the present field direction towards the ESE-down direction but upon further demagnetization their paths trace out a clockwise-spiral pattern with the overall tendency to approach that of the typical Sudbury dike direction. These rocks record three components of magnetization in overlapping coercivity spectra with the Sudbury dike direction predominantly in the hardest part of the spectrum. The Sudbury dike direction is defined with very low within-site precision because it is not reached until the magnetization is reduced to a few percent or less of its initial intensity at which point the true direction is obscured by imperfectly demagnetized softer components.

These sites: 11, 20, 41, and 83 are located near the Grenville Front. West of these sites, Sudbury dike directions prevail as the single isolated direction of remanence; east of these sites, the ESE-down direction prevails as the single characteristic remanence direction. This spatial distribution suggests that some or possibly all of the ESE-down directions of magnetization are the result of a remagnetization process whereby the original thermal remanent magnetization (TRM) (Sudbury dike direction) has been completely replaced by a younger magnetization related to the 1000 Ma thermal event recorded throughout the Grenville Province.

Baked Contact Studies

At a number of sites the contact between the dike and the country rock is exposed. At these sites samples of the country rocks were taken from the baked zone and up to several dike-widths away. Most of these tests proved to be inconclusive inasmuch as many of the Huronian country rocks sampled are either clean quartzites (Lorrain Formation) or limestones of the Espanola Formation, both of which possess a weak and generally unstable remanence. The most satisfactory country rocks proved to be argillites and siltstones of the Gowganda Formation. Site 90 (Fig. 1) was located in the Cobalt Plate where a Sudbury diabase dike having the Sudbury dike direction of magnetization is in intrusive contact with argillites of the Gowganda Formation. Two sites were located (23 and 72 on Fig. 2) where ESE-down magnetized dikes are in contact with Gowganda argillites and siltstones.

The results of the baked contact test at site 90 are displayed in Fig. 9. Although we were unable to sample completely unbaked Gowganda material in the vicinity of site 90 the characteristic regional remanent magnetization direction of the Gowganda Formation is steeply down to the north (Symons 1975; Morris 1977). Three cores within the baked zone of the diabase are close to, but not in excellent agreement with, the baking diabase. Two of the three cores of the baked argillites are not very stable; they exhibited erratic directional changes after 400 Oe (31.8×10^3 A/m) and 500 Oe (39.8×10^3 A/m) AF treatment whereas the four cores from the diabase retained directional coherency up to 700 Oe (55.7×10^3 A/m) alternating magnetic fields. This result, coupled with the excellent contact tests of Sudbury diabases against Sudbury norite presented by Sopher (1963) and Schwarz (1977), provide strong evidence that the Sudbury dike magnetization is primary.

At site 72, an ESE-down magnetized dike near the Grenville Front, quite different results were obtained. A single characteristic remanence direction was isolated from the dike at moderate AF fields (Fig. 10). The argillites, collected at intervals ranging from 0.05 dike width to 2 dike widths from the intrusive contact, are magnetically harder than the dike, but all cores, regardless of their distance from the dike, possess an ESE-down direction of magnetization. This magnetization is independent of dike intrusion and we interpret this result to indicate that both the dike and the country rocks have been remagnetized together.

Two dikes or dike segments exposed to the north of McGregor Bay and sampled as sites 10, 23–27, and 65 (see Fig. 2) also possess the ESE-

TABLE 2. Site magnetic properties

Site	Specimens	J_0 (emu/g × 10^{-5})	m.d.f. NRM (Oe)	$Q = \frac{J_0}{\chi H}$	T_c (°C)
1	7	72.8 ± 40.5	56 ± 22	2.0 ± 1.0	—
2	8	29.4 ± 8.4	146 ± 28	0.42 ± .12	—
3	6	44.2 ± 17.1	103 ± 14	0.68 ± .16	—
5	3	54.8 ± 17.9	125 ± 18	0.76 ± .07	—
6	6	38.0 ± 1.8	204 ± 30	0.97 ± .09	—
8	6	32.5 ± 2.7	174 ± 19	0.83 ± .13	—
9	7	82.6 ± 59.3	144 ± 72	2.04 ± 1.46	555
10	7	91.4 ± 21.8	115 ± 88	1.85 ± .58	565
11	7	24.2 ± 7.8	61 ± 5	0.60 ± .27	565
13	5	162. ± 86.1	123 ± 114	2.25 ± .81	—
14	3	90.5 ± 37.3	32 ± 11	0.96 ± .50	—
16	5	64.6 ± 12.5	133 ± 12	1.69 ± .23	—
17	7	136. ± 60.2	46 ± 12	1.03 ± .48	580
18	6	42.2 ± 15.2	39 ± 14	0.72 ± .18	567
19	8	231. ± 73.8	30 ± 3	3.18 ± .96	570
20	8	76.8 ± 14.0	36 ± 6	1.32 ± .21	—
21	8	35.6 ± 1.9	116 ± 12	0.85 ± .10	—
22	7	18.2 ± 2.0	247 ± 10	0.55 ± .12	555
23	8	56.4 ± 10.8	240 ± 12	1.02 ± .22	565
24	8	296. ± 53.7	239 ± 20	5.03 ± .89	582
25	8	87.3 ± 10.2	199 ± 32	1.27 ± .20	555
26	8	52.8 ± 3.6	276 ± 12	1.06 ± .02	—
27	8	51.3 ± 4.1	150 ± 26	0.95 ± .06	—
28	8	80.4 ± 13.9	206 ± 114	1.44 ± .30	—
29	7	31.4 ± 12.6	191 ± 32	0.50 ± .14	—
31*	7	874. ± 203.	63 ± 13	16.5 ± 4.01	—
32*	8	730. ± 306.	50 ± 13	12.6 ± 5.30	—
33*	7	5610. ± 1420.	83 ± 4	119. ± 29.2	—
34	8	410. ± 397.	42 ± 6	5.72 ± 5.56	—
35	8	187. ± 39.1	36 ± 8	3.44 ± .85	570
36	8	1380. ± 1650.	73 ± 18	19.9 ± 22.8	—
37	8	124. ± 13.9	31 ± 3	2.09 ± .18	575
38	7	2120. ± 3800.	128 ± 27	47.1 ± 85.5	—
40	8	228. ± 114.	89 ± 27	3.37 ± 1.71	570
41	6	23.4 ± 10.7	75 ± 13	0.46 ± .19	570
43	7	105. ± 135.	50 ± 14	2.00 ± 2.55	—
44*	7	6880. ± 3290.	76 ± 16	131. ± 67.9	—
45*	8	3080. ± 4360.	45 ± 22	68.3 ± 107.	—
46	7	51.5 ± 11.9	145 ± 26	1.16 ± .28	—
47	8	50.8 ± 21.2	99 ± 38	1.05 ± .47	—
48*	8	1120. ± 1080.	48 ± 8	15.2 ± 13.9	—
49	8	104. ± 140.	52 ± 20	1.66 ± 2.50	—
50	8	59.4 ± 5.5	146 ± 11	0.73 ± .07	—
51	5	38.0 ± 5.3	159 ± 21	0.86 ± .10	—
52	8	84.9 ± 30.6	114 ± 57	3.25 ± 3.50	—
53	6	195. ± 379.	133 ± 56	3.39 ± 5.50	—
54	8	31.6 ± 2.3	156 ± 15	0.80 ± .11	—
55	8	26.9 ± 6.1	102 ± 49	0.45 ± .02	—
56	7	55.4 ± 25.2	123 ± 57	0.69 ± .11	—
57	7	20.0 ± 4.2	179 ± 20	0.57 ± .05	—
58	8	33.9 ± 9.7	179 ± 30	0.74 ± .18	—
59	8	338. ± 279.	68 ± 49	7.43 ± 7.37	—
60	8	51.0 ± 14.9	150 ± 18	0.72 ± .08	—
61	8	93.6 ± 65.2	92 ± 48	2.60 ± 3.65	—
62	8	40.4 ± 9.0	43 ± 14	0.52 ± .12	—
63	8	296. ± 38.3	110 ± 13	3.50 ± .18	—
65	4	19.3 ± 2.3	145 ± 41	0.52 ± .13	—
66	4	58.3 ± 11.6	105 ± 6	1.03 ± .19	—

TABLE 2. (*Concluded*)

Site	Specimens	J_0 (emu/g × 10^{-5})	m.d.f. NRM (Oe)	$Q = \frac{J_0}{\chi H}$	T_c (°C)
68	6	46.7 ± 11.2	98 ± 23	1.09 ± .31	540
69	4	37.5 ± 10.5	89 ± 20	0.81 ± .24	540
70a*	5	3450. ± 5020.	60 ± 10	34.0 ± 30.4	—
70b*	3	3010. ± 3830.	54 ± 12	29.9 ± 34.6	—
72	4	77.3 ± 34.4	33 ± 5	0.96 ± .64	—
73	4	93.9 ± 45.8	35 ± 1	1.07 ± .36	—
74	3	37.8 ± 8.8	46 ± 1	0.76 ± .06	560
75	4	28.9 ± 7.2	46 ± 10	0.54 ± .06	560
76	4	59.5 ± 56.4	66 ± 35	2.28 ± 3.23	—
77	3	147. ± 111.	34 ± 9	4.65 ± 3.37	—
79	4	62.2 ± 14.5	90 ± 3	0.60 ± .07	—
80	4	55.6 ± 2.5	109 ± 68	0.68 ± .04	—
83	6	33.5 ± 2.0	135 ± 4	0.63 ± .06	—
84	8	52.8 ± 7.2	232 ± 20	1.26 ± .19	—
85	8	73.5 ± 56.0	60 ± 11	1.51 ± 1.18	—
48r	4	197. ± 251.	37 ± 3	2.72 ± 3.22	—
88*	7	8570. ± 456.	48 ± 12	100. ± 16.7	—
89	5	10.8 ± 4.0	62 ± 20	1.97 ± 1.51	—
90	8	39.9 ± 13.0	246 ± 43	1.11 ± .44	—
91	8	149. ± 39.4	299 ± 9	1.87 ± .46	—
92	6	102. ± 27.0	198 ± 98	1.12 ± .26	—
93	7	66.4 ± 27.5	199 ± 19	1.57 ± 1.24	—

NOTES: The intensity of NRM in cgs emu units may be obtained by multiplying the values of column 3 by the density which for these diabases is 2.9 g/cm³.
In calculating the Koenigsberger ratios of column 5, a field value of 0.6 Oe was used.
The Curie temperatures were measured in an 8 kG field.
The m.d.f. is the median destructive field, the alternating field that reduces the NRM to one half.

down direction of magnetization yet are a considerable distance from the Grenville Front. If the remagnetization process is the sole origin for the ESE-down direction of magnetization it would imply that the zone of remagnetization has a very variable width in advance of the Grenville Front. Alternatively, some of the ESE-down magnetized dikes may represent geologically younger dikes intruded during the episode of remagnetization. The magnetic evidence at sites 23 and 24 supports the second interpretation. At this locality a 40-m wide dike is exposed together with a 5-cm wide dikelet both of which are intruded into Gowganda argillites. Four cores were taken from the dikelet (site 24), four cores from the main dike (site 23), and four cores from the argillites at varying distances from the site 23 dike contact. The results of this baked contact study are presented in Figs. 11 and 12. Both the intensity and the magnetic stability of the Gowganda argillites are increased in the baked zone. Furthermore, the cores of the argillite in the baked zone are in good directional agreement with both the baking dike and the associated dikelet whereas the single core distant from the baked zone is much less directionally stable, possessing a northerly and down direction of magnetization over most of its coercivity spectrum. This northerly and down direction of magnetization is common to all of the unbaked Gowganda argillites in the Whitefish Falls area (W. A. Morris, personal communication, 1976).

K–Ar Dating and Whole Rock Petrochemistry

In the previous section evidence was presented which indicated that the remanence of the dikes in or near the Grenville Province was reset during the last Grenville metamorphic event. The boundary of the area where metamorphism was severe enough to have reset the TRM is defined by the location of the 'spiralling' paleomagnetic sites: 11, 20, and 41 of Fig. 2 and site 83 of Fig. 5. Rubidium–strontium dating of the dikes within the Southern Province yielded ages, believed to be intrusive, of 1205–1460 Ma. The question then arises, how did the Grenvillian event affect the apparent ages of the dikes near and within the Grenville Province? It is to be expected that within the Grenville Province K–Ar dating would yield metamorphic ages of about 900–1100

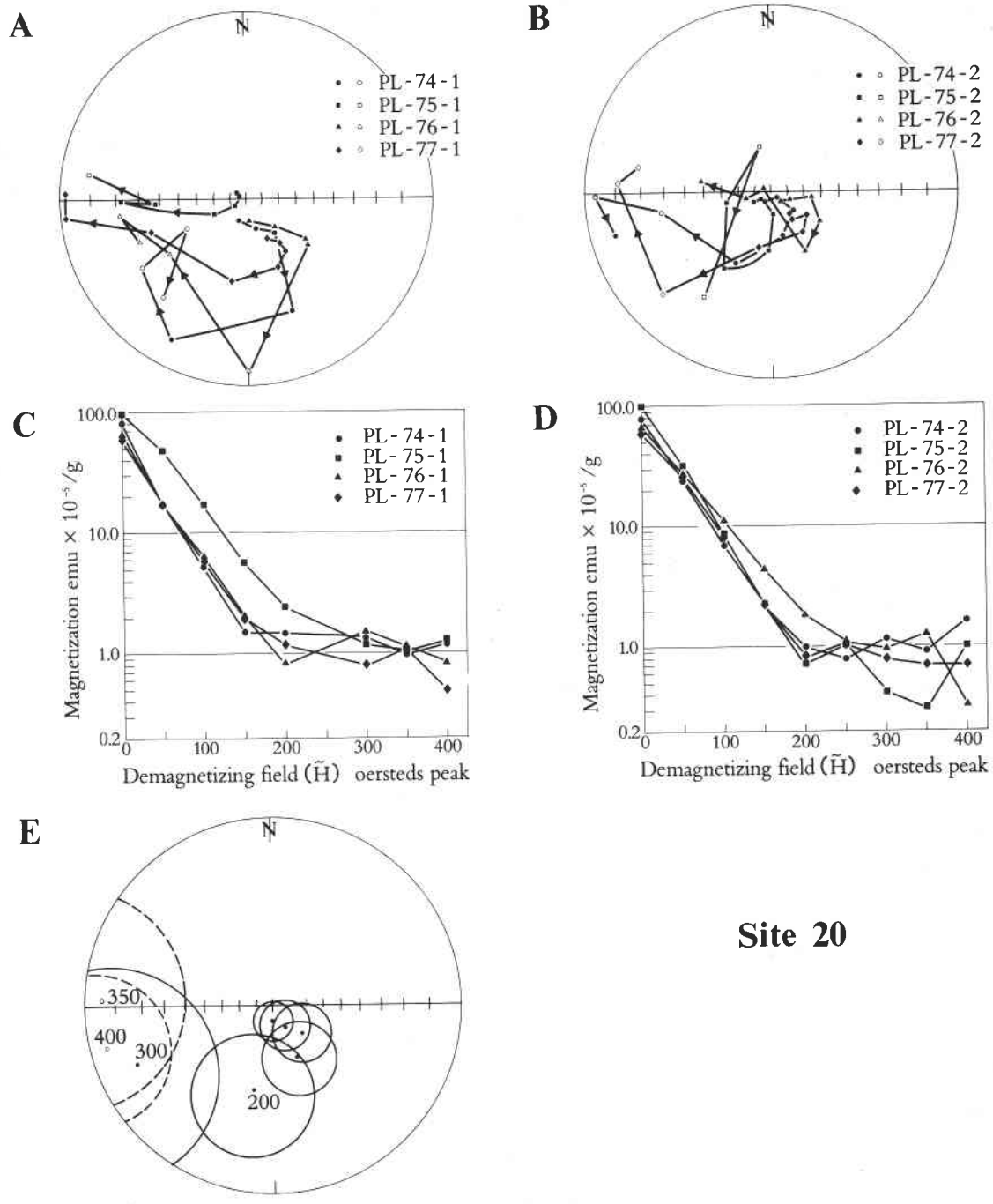


FIG. 7. Behaviour of site 20 samples ('U' type site from near the Grenville Front) with AF demagnetization. Directional changes and intensities of remanence with progressive AF demagnetization up to 400 Oe (31.8×10^3 A/m) peak for first specimens are shown in A and C. Second specimen results are shown in B and D. Specimen 1 and specimen 2 directional data were first combined to yield sample means which when combined yield the overall site means and their associated circles of 95% confidence as plotted in E. Closed (open) symbols indicate north-seeking directions on the lower (upper) hemisphere of the equal area projections of A, B, and E.

Ma whereas within the Southern Province intrusive ages of 1200–1400 Ma should be found. The dikes to the north of McGregor Bay (sampled at sites: 10, 23–27, and 65) are believed, from paleomagnetic baked contact evidence, to be of Grenvillian age and therefore should have K–Ar ages similar to the dikes within the Grenville Province. The primary purpose of the dating survey was to determine where, in relation to the location of the transition from original to reset TRM, as marked by the 'spiralling sites', a transition from intrusive to metamorphic apparent K–Ar ages occurred.

Samples from 27 sites of Fig. 2, from site 60 in region A of Fig. 1, and from site 80 of Fig. 3 were dated by the K–Ar whole-rock method. Samples were crushed to -20 $+100$ mesh, homogenized, and split into four portions. One portion was then crushed to -100 mesh for potassium analysis, which was by the flame photometric method using an Instrumentation Laboratory flame photometer, Model 143. A lithium internal standard and a sodium buffer were used. Argon analysis was by the isotope dilution method and an AEI MS-10 mass spectrometer was employed. The results are given in Table 3.

The results show that, although Grenvillian ages were found within the Grenville Province, the Southern Province apparent ages were unexpectedly scattered. The existence of any transition from intrusive to metamorphic ages was masked by a zone of anomalously old apparent ages in the region of the 'spiralling' paleomagnetic sites. The 1542 Ma, 1630 Ma, and 1700 Ma, and older ages found at sites 20, 40, and 41 are anomalous. The fact that the dikes intrude the Killarney Granite which has been dated by Rb–Sr at 1571 Ma $\pm$ 47 Ma (Wanless and Loveridge 1972) and by U–Pb at 1550 Ma (Krogh *et al.* 1971) shows that these high apparent ages are not meaningful in a geochronological sense. Stockwell (1963, pp. 129–130) first pointed out that anomalously old K–Ar biotite ages are occasionally found in the Grenville Front region. Subsequently, both Wanless *et al.* (1970), who performed a dating survey across the Grenville Front near Chibougaumau, Quebec, and Giletti (1971) who worked in Montana, have found anomalously old apparent K–Ar ages on biotites from rocks adjacent to metamorphic terrains. Giletti ascribes the anomalous ages to excess argon gained by the biotites during the regional metamorphism of the adjacent area. Gases liberated from the hottest parts of the metamorphic terrain could be absorbed by the cooler rocks on the margins of the terrain. Thus the Grenville metamorphic event could have been the cause of the anomalous ages found at sites 20, 40, and 41.

At sites 68 and 69 unexpectedly young, 401 Ma and 431 Ma, ages were found. Two samples, provided by the Geological Survey of Canada, from outcrops near sites 11 and 46 were dated at 425 Ma and 492 Ma respectively, confirming similar results obtained previously by that organization.

Chemical analyses, by X-ray fluorescence, were performed to look for significant compositional differences between the dikes believed to be intruded at different times; i.e. the majority of the dikes intruded at about 1200–1400 Ma, the McGregor Bay dikes dated at sites 24 and 27 at 1020 Ma and 1044 Ma and believed to be genuinely Grenvillian, and the dike or dike segments characterized by sites 68 and 69 with early Paleozoic apparent ages.

A Phillips Automatic Sequential Spectrometer, PW 1450, was used to analyse glasses, prepared from the rock powders made during the K–Ar work, for the major elements: Si, Al, Fe, Mg, K, and S. Pressed pellets, prepared from the rock powders, were analysed for Na, Ti, and the trace elements: Cr, Ba, Ni, Nb, Zr, Y, Sr, and Rb. The complete analytical data are given in Merz (1976).

Figure 13 illustrates the principal findings of the petrochemical study. Figures 13a and 13b show Ba and TiO_2 plotted against Miyashiro's (1975) fractionation index; it can be seen that the samples with early Paleozoic apparent ages lie on a different fractionation trend from the other dike samples. The alkali–silica plot, Fig. 13c, shows that the young samples are tholeiites whereas the others are alkali olivine basalts. No signs of remobilization of alkalis are seen in the thin sections of the samples so this too is evidence that the young samples represent a distinct episode of intrusion. On geochemical grounds alone, the McGregor Bay dikes are not distinct from the Sudbury dikes possessing either the 'Sudbury dike direction' or the 'ESE-down direction' of magnetization.

Geology and Petrography

The average width of the dikes of this study is

approximately 30 m but the observed range in outcrop width varies from 0.05 m (off-shoots of larger dikes) to about 150 m. Only in a few outcrops can the full width of a dike exposure be found and outcrops with dike – country rock contacts exposed are not abundant. Lineaments, seen on air photographs and aeromagnetic maps, have been used by the authors of the geological maps upon which Figs. 1–5 of this report are based, to help delineate the position of dikes between outcrops of the diabase. In the Southern Province – Cobalt Plate, outcrops of the Sudbury diabase can be distinguished from the more abundant Nipissing diabase by a more rusty appearance of the former type and by the fact that in the Southern Province the Nipissing diabase is a metadiabase with most of its original pyroxene now converted to an actinolitic amphibole. This mineralogical feature is most obvious during field-drilling as the Nipissing diabase yields a green coloured drilling mud whereas the Sudbury diabase produces one of grey colour. Well into the Grenville Province where metamorphism has affected both the Nipissing diabase and the Sudbury diabase, the distinction between the two is less unequivocal.

Our petrographic observations of the diabases will be grouped into the following categories: (1) Sudbury diabases of the Southern Province including the McGregor Bay dikes; (2) Sudbury diabases in the Grenville Front region; (3) the metadiabases well within the Grenville Province, the locations of which are shown on Fig. 3b; (4) The diabases in the Grenville Front region with early Paleozoic apparent ages.

Sudbury Diabases of the Southern Province including the McGregor Bay Dikes

These dike rocks usually display an ophitic to subophitic texture and consist of the following three major primary minerals in descending order of abundance: plagioclase of intermediate composition, titaniferous augite (1.5–2.0% TiO_2 as determined by microprobe analysis), and olivine. Primary biotite, commonly in association with Fe–Ti oxides, is invariably present and lesser amounts of accessory apatite are commonly present. The abundant olivine and the minor, but ever present, biotite are the mineralogical expressions of the alkali olivine basalt chemistry of these rocks. Many of the slides examined show no evidence of late stage deuteric alteration or later regional metamorphism; many of the olivines are completely unserpentinized. Where there is evidence of secondary alteration this most commonly is the partial or complete transformation of olivines to serpentine–chlorite–oxide minerals or to serpentine–talc–oxide minerals. Sericitization and (or) kaolinization of plagioclase is present sporadically as is the development of small amounts of chlorite. Uralitic alteration of the clinopyroxenes is very rare.

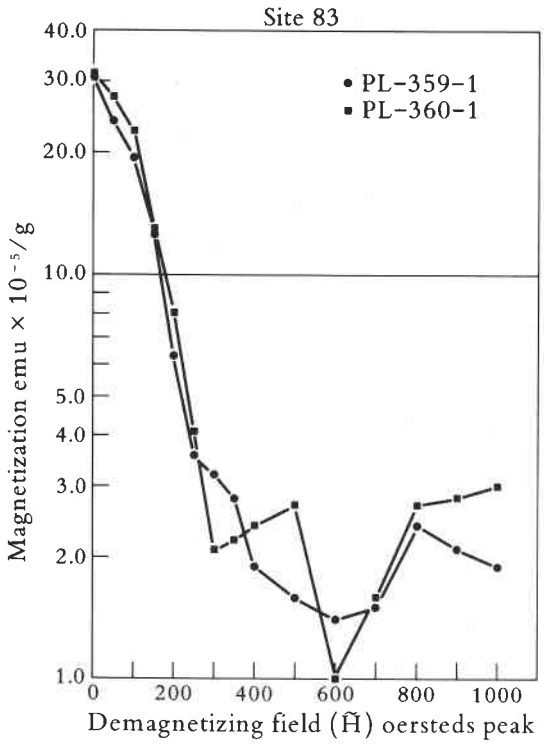
The Fe–Ti oxides consist of rather abundant separate ilmenite grains and titaniferous magnetite grains. The titaniferous magnetites are not single phases but have included ilmenite lamellae, the abundance of which places them in oxidation classes 2 and 3 of Ade-Hall *et al.* (1968). Sites possessing median destructive fields of NRM in excess of 100 Oe (7.96×10^3 A/m) (see Table 2) have their titaniferous magnetites further subdivided into magnetic and nonmagnetic regions by class 5 spinel rods or more commonly by the development of magnetite–ulvospinel intergrowths in the regions between larger ilmenite lamellae. Sulfides are much less abundant and include pyrrhotite, pentlandite, chalcopyrite, possibly cubanite, and rarely pyrite.

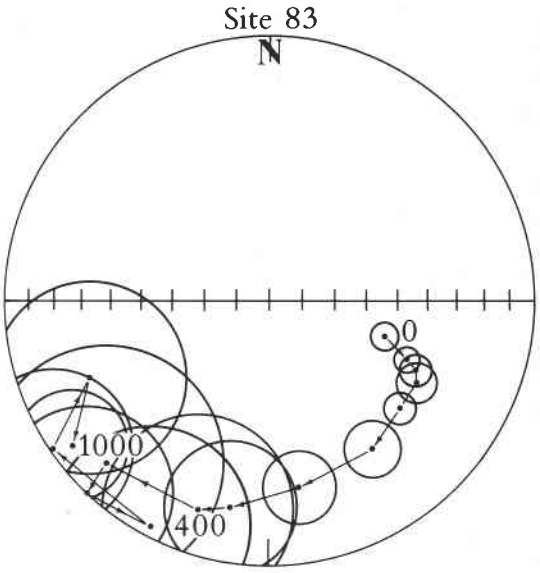
Sudbury Diabases in the Grenville Front Region

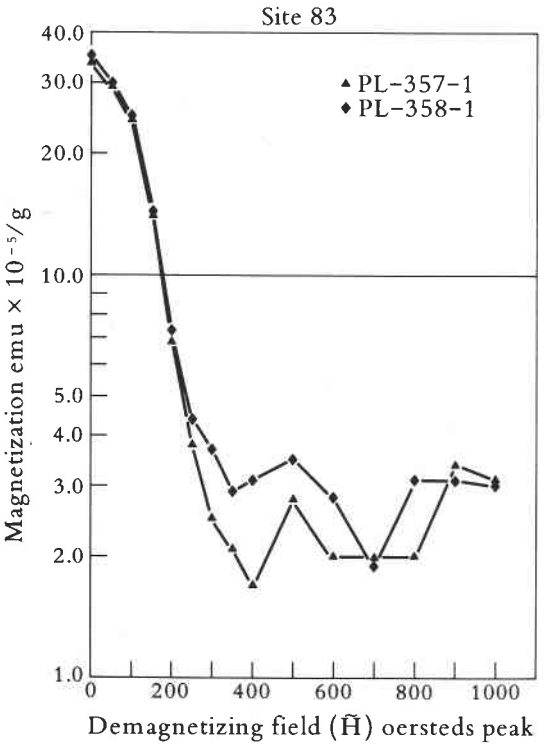
This subsection applies to the southeast corner of Fig. 2 and sites: 61, 62, and 63 of subregion H depicted on Fig. 1. In addition this section applies to the 'spiralling' sites: 11, 20, and 41 immediately in advance of the Grenville Front. Unclassified sites (site numbers suffixed with the letter U) on Fig. 2 in this region are also included.

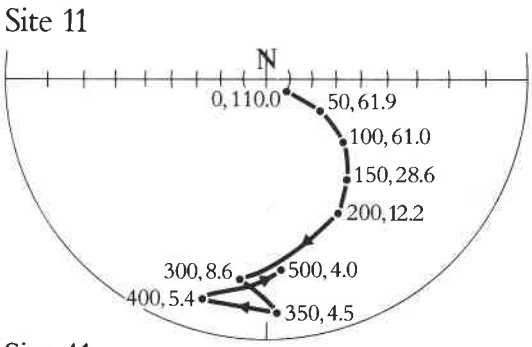
Briefly stated, all sites possessing the spiralling

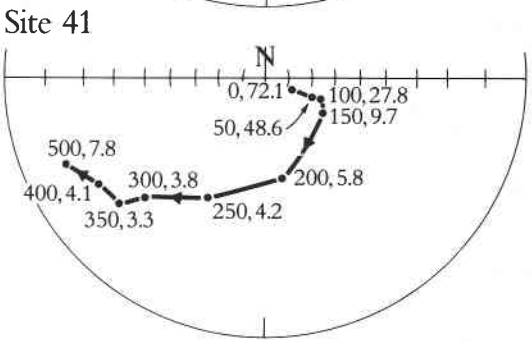
FIG. 8. Demagnetization curves for four samples from site 83 are shown on the left of the figure; the directional behaviour for this site is summarized in the equal area projection where the site-mean directions are plotted with their associated circles of confidence. For sites 11 and 41, which are also 'U' type sites near the Grenville Front, the directional data only are displayed. The AF field at which the mean direction was obtained and the value of k for $N = 4$ are given beside each direction. Site 11 shows only partial movement toward the 'Sudbury-type direction' which is believed to be the original TRM direction.


Site 83
• PL-359-1
▪ PL-360-1
Magnetization emu × 10⁻⁵/g
40.0
30.0
20.0
10.0
5.0
4.0
3.0
2.0
1.0
0
200
400
600
800
1000
Demagnetizing field (H̃) oersteds peak


Site 83
N
0
1000
400


Site 83
▲ PL-357-1
♦ PL-358-1
Magnetization emu × 10⁻⁵/g
40.0
30.0
20.0
10.0
5.0
4.0
3.0
2.0
1.0
0
200
400
600
800
1000
Demagnetizing field (H̃) oersteds peak


Site 11
N
0, 110.0
50, 61.9
100, 61.0
150, 28.6
200, 12.2
300, 8.6
500, 4.0
400, 5.4
350, 4.5


Site 41
N
0, 72.1
100, 27.8
150, 9.7
50, 48.6
500, 7.8
300, 3.8
200, 5.8
400, 4.1
350, 3.3
250, 4.2

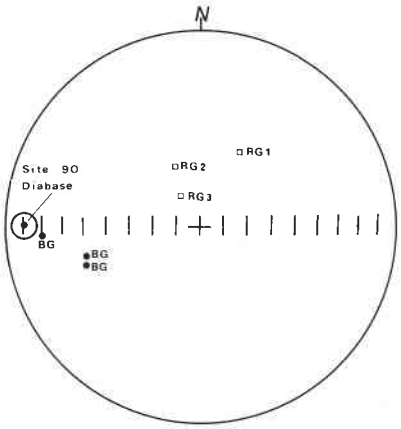


FIG. 9. On the equal area plot are shown the mean direction and its circle of 95% confidence of four cores from the diabase at site 90 which is a representative site having the 'Sudbury dike direction'. BG are the directions obtained from 3 cores of baked Gowganda argillites collected within 0.5 dike-widths of the diabase dike. RG 1 and RG 2 are estimates of the primary and early secondary regional directions from the Gowganda Formation in the Cobalt Plate as reported by Morris (1977). RG 3 is the regional direction from the Gowganda Formation in the Elliot Lake region as reported by Symons (1975).

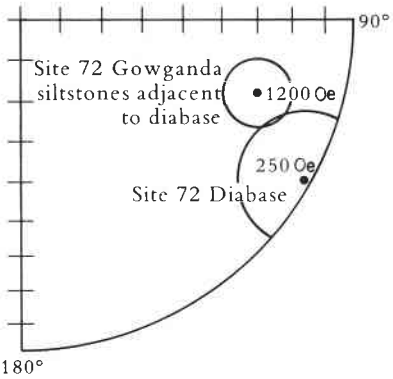


FIG. 10. On a portion of an equal area plot are shown the mean direction and its circle of 95% confidence of four cores from the diabase at site 72. Also plotted is the mean of 4 cores of Gowganda Formation argillites collected at 0.05 dike-width to 2 dike-widths from the site 72 dike.

paleomagnetic signature or which possess the totally reset ESE-down magnetization differ in no significant respect from those of the Southern Province described above, i.e. they in no sense can be described as metamorphic rocks. These apparently unmetamorphosed sites occur in close spatial association with sites which do exhibit cataclastic textures and (or) thoroughly reconstituted mineralogy. Brooks (1976) has described one such dike locality (our site 88) where the diabase has been reconstituted to a min-

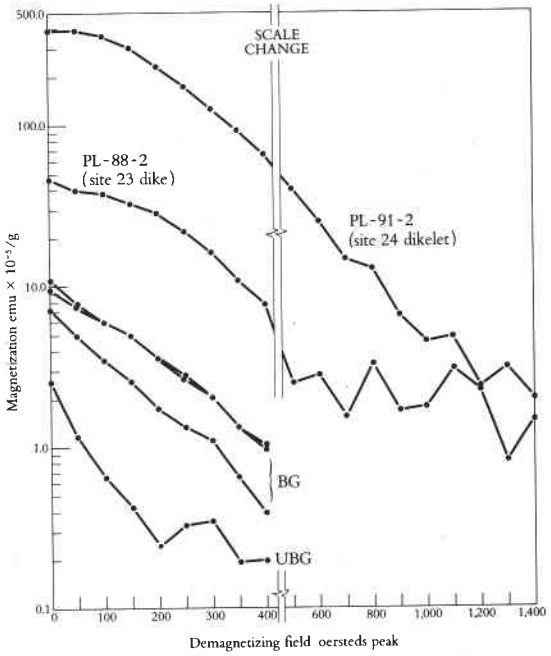


FIG. 11. Demagnetization curves for 1 specimen from each of two dikes at sites 23 and 24 together with the demagnetization curves of the four specimens of the Gowganda argillites adjacent to site 23. BG and UBG refer to baked Gowganda and unbaked Gowganda respectively.

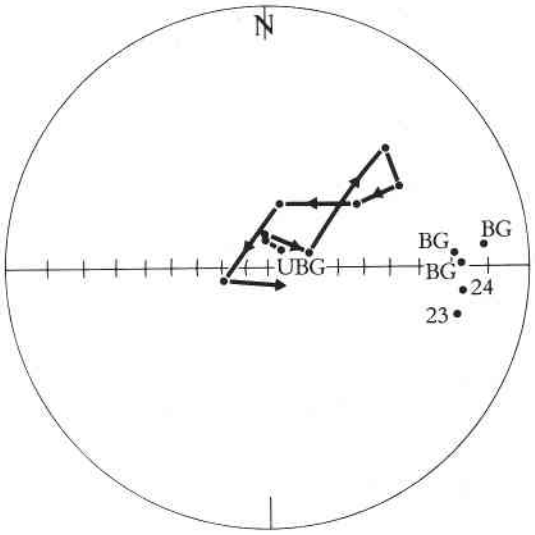


FIG. 12. On the equal area projection are plotted the site-mean directions for sites 23 and 24 at 300 Oe (23.9×10^3 A/m) and 250 Oe (19.9×10^3 A/m) respectively. The dots labelled BG are the directions obtained at 400 Oe (31.8×10^3 A/m) from 3 baked Gowganda cores collected within 2 m of the site 23 dike. The series of points labelled UBG are the directions obtained at successive stages of demagnetization from the one unbaked Gowganda core collected 1 dike-width away from the contact.

TABLE 3. K–Ar data

Site no.	Sample no.	K (%)	Ar_{rad}^{40} $\times 10^{-5}$ ml g^{-1}	Calculated age, Ma
near 11	FC-BL-50-65	0.342	0.653	425
near 46	FAD-410-65	0.229	0.515	492
68	PL 281	0.347	0.671	431
69	PL 284	0.464	0.893	429
69	PL 286	0.714	1.273	401
2	PL 6	1.23	11.25	1503
8	PL 28	1.21	9.365	1337
9	PL 31	0.873	6.363	1277
11	PL 39	0.994	8.455	1426
14	PL 52	1.05	6.286	1106
16	PL 59	0.934	3.859	825
17	PL 62	0.960	6.528	1214
19	PL 73	1.19	6.499	1029
20	PL 76	1.08	10.31	1542
22	PL 83	0.638	4.320	1210
24	WF 11	1.21	6.557	1020
27	PL 102	1.495	8.327	1044
29	PL 111	1.05	7.786	1295
34	PL 131	1.20	7.510	1142
36	PL 141	1.36	8.845	1174
38	PL 147	0.902	7.134	1354
40	PL 157	0.694	7.176	1630
41	PL 158	0.947	11.02	1762
41	PL 288	0.929	11.55	1838
45	PL 177	1.255	10.55	1414
52	PL 203	1.73	8.996	990
54	PL 217	1.36	7.637	1049
58	PL 242	0.991	10.24	1629
60	PL 251	1.08	6.937	1162
70	PL 296	1.065	5.646	1005
76	PL 323	1.00	5.134	980
80	PL 340	1.15	7.231	1144
near 90	GSC-64-100*			1240

NOTES: *This entry is that reported by Wanless *et al.* (1966). The determination was made on a biotite concentrate separated from diabase near our site 90.

eralogical assemblage appropriate to the greenschist facies. Our site 13 is cataclastically deformed but without much neomineralization; actinolite and chlorite have been produced but some original clinopyroxene still remains. Our site 15 (not plotted on Fig. 2) was originally dismissed as Nipissing diabase because it occurs in close proximity to essentially fresh Sudbury diabase at site 16. On re-examination it is a biotite-bearing amphibolite and is probably Sudbury diabase rather than Nipissing diabase. The point we make is that Sudbury diabase with virtually unaltered igneous mineralogy and metamorphosed Sudbury diabase with mineralogy approaching that of the amphibolite facies occur in close spatial association in this region. Those dikes or dike segments which have widths less than 10 m seem to have acquired the necessary water (from adjacent country rocks?) at the time of their metamorphism to have achieved mineralogical expression of the Grenville heating event. The exposed central portions of wider dikes, although reheated and reset magnetically, were presumably armoured from mineralogical expression of this reheating by the unavailability of water.

Metadiabases of Subregion C of Fig. 1 (Sites 74–77 of Fig. 3b)

Sites 74–77 are in small bodies of recrystallized and cataclastic metadiabase exposed along Highway 637 (see Fig. 3b). Although the bodies are not dike-like in form they may represent disrupted portions of dikes; in any event, the microscopic evidence is consistent with their being alkali olivine diabases of the Sudbury diabase magma type. Although weakly foliated they exhibit relict subophitic and intersertal microtextures with abundant biotite concentrated about relict ilmenite accessories. These meta-

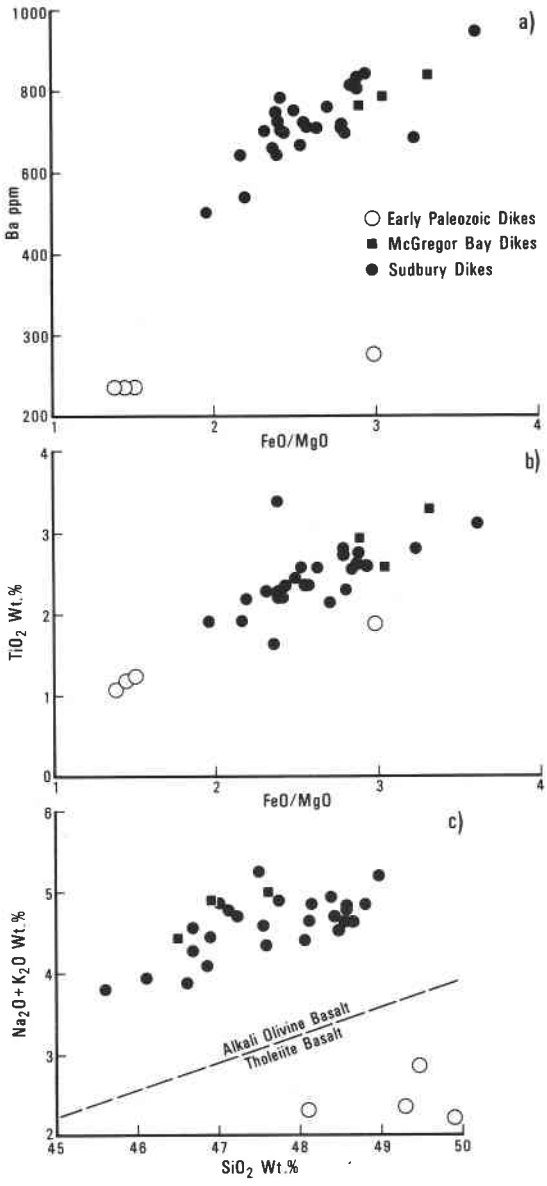


FIG. 13. Chemical variation diagrams for some of the dike samples. In (a) and (b), FeO is total iron calculated as FeO. MacDonald and Katsura's (1964) boundary between tholeiite and alkali olivine basalt series for Hawaiian volcanic rocks is shown as a reference line on part c.

diabases consist of plagioclase, pyroxene, biotite, garnet, possibly spinel, and iron–titanium minerals. The plagioclase occurs as recrystallized aggregates and relict primary laths having intensely clouded cores displaying oscillatory zoning. These relict cores are bent and fractured. Although relict cores of clinopyroxenes occur, they differ from those in the diabases of the Southern Province and Grenville Front region in having lower birefringence and being clouded by numerous rod-shaped inclusions of Fe–Ti oxide minerals. Many of the cores of clinopyroxene are surrounded by corona-like structures of clear clinopyroxene, orthopyroxene, biotite, and garnet. Some of the smaller isotropic grains may be spinels. These rocks appear to have formed anhydrous granulite facies assemblages without having first passed through the intervening greenschist and amphibolite facies; we reach this conclusion because vestiges of the original diabasic texture remain, whereas the thin diabases of the Grenville Front region which have been converted to amphibolite dikes have their primary igneous textures destroyed.

The demagnetization curves of these rocks differ from the diabases of the Southern Province and Grenville Front region. Although the median destructive fields are low (Table 2) the remaining 30–40% of the remanence is hard. This hard remanence presumably resides in the tiny rods of iron–titanium oxides in the pyroxene relicts and plagioclase cores.

Diabases in the Grenville Front Region with Early Paleozoic Apparent Ages

Our observations on this type of diabase are restricted to samples from our sites 46, 68, and 69 together with an examination of sample FC-BL-50-65 supplied by M. J. Frarey and collected near our site 11. These samples are distributed along a 4 km strike length of a longer, apparently continuous dike. We suppose that Sudbury dike material and this younger, chemically distinct magma type occupy the same west-northwest fracture system. We have not discovered any reliable field criteria to distinguish the two types. Microscopically however, the following features are characteristic of the early Paleozoic dike material:

(1) Biotite is virtually absent from these rocks whereas it constitutes one or two percent of the Sudbury diabases.

(2) Olivines in the younger dikes are almost wholly serpentinized; olivines in the Sudbury diabases are commonly fresh.

(3) Clinopyroxenes are colorless whereas they are usually pink titanaugites in the Sudbury diabases.

(4) Separate ilmenite grains are rare in the younger diabases.

(5) Although titaniferous magnetites are sub-

divided by ilmenite lamellae producing class 2 and 3 type grains, the residual magnetite is not homogeneous but is broken up by a patchy microtexture into magnetite areas and into areas having much lower reflectivity. In some titaniferous magnetites, the residual areas between ilmenite lamellae have undergone 'granulation' of the type described by Ade-Hall *et al.* (1968). High field Curie temperatures in the young diabases are somewhat lower than in all other samples (see entries for sites 68 and 69 in Table 2).

Conclusions and Discussion

The direction of magnetization obtained and the calculated paleomagnetic pole resulting from that direction is little different from that of previous studies of the Sudbury diabases exposed in the Southern Province. A pole of high accuracy (168° W, $2\frac{1}{2}$° S, $A_{95} = 2\frac{1}{2}$°) results from the combination of the results of this study with those of Larochelle (1967). The paleomagnetic and petrochemical data of this study serve to extend the appellation 'Sudbury diabase' to a larger area including the Cobalt Plate. These dike rocks have alkali olivine basalt chemistry which distinguishes them from most other diabase dike swarms of the Canadian Shield (Fahrig *et al.* 1965). Although the magnetization of the Sudbury diabases in the Southern Province – Cobalt Plate is undoubtedly primary, the age of the dikes carrying this magnetization is less well defined than one would hope. Van Schmus (1975) has given convincing arguments for adopting an age of 1250 Ma $\pm$ 50 Ma as the current best estimate. A mean whole-rock K–Ar age close to this estimate can be made from the data of Table 3 by excluding those sites within and near the Grenville Province and the sites from the McGregor Bay dikes carrying the ESE-down magnetization. However, the scatter about this mean is large. As examples, sites 2 and 58 yield apparent ages which are improbably old; sites 52 and 54 yield apparent ages which are improbably young. We make these statements solely on the basis of current estimates of the North American Precambrian polar wander curve and will be the first to acknowledge the circular reasoning of this argument. Explanations for the presumed loss of argon in the younger sites and presumed excess argon in the older sites were looked for; no correlation of apparent age of site with either petrography or the nature of its enclosing country rocks was found.

The most important conclusions are those pertaining to the Grenville Front region.

(1) The original magnetization (Sudbury dike direction) and a reset magnetization (ESE-down) coexist in the same samples over a very short distance—probably less than 1 km. This belt of partial remagnetization occurs about 2 km in advance of the Bell Lake granite and about 4 km in advance of the Grenville Front as drawn by Card (1971*a*). Near the Quebec border the position of resetting is about 8 km in advance of the Grenville Front (site 83 of Fig. 5).

(2) In this belt of partial magnetic resetting, whole-rock K–Ar ages are markedly increased presumably by the incorporation of excess argon at the time of reheating and remagnetization.

(3) Total remagnetization takes place without any change in petrography and without any adjustment in high field Curie temperatures (Table 2). The most likely mechanism of remagnetization is that discussed by Pullaiah *et al.* (1975) whereby original high temperature TRM is unblocked by heating to moderate temperatures if the heating time is geologically long. By this mechanism igneous rocks may possess a metamorphic magnetization.

(4) The reheating temperature in the Grenville Front region was probably of the order of 400°C. This estimate is based upon the sporadic transformation of some thin dikes to metamorphic mineral assemblages of the amphibolite facies. The amphibolite dikes themselves do not possess a stable magnetization.

(5) The time of remagnetization is estimated at about 1.0 Ga. This estimate is based upon whole-rock ages from sites: 14, 16, 17, 19, and 76 in the Grenville Front region and from whole-rock ages from sites 24 and 27. Sites 24 and 27 are in the McGregor Bay dikes which we believe possess a primary magnetization dating from the time of intrusion. Thus sites possessing the ESE-down direction of magnetization are in dikes of two origins; remagnetized Sudbury diabases of age 1.2–1.3 Ga and chemically similar dikes intruded during the period of remagnetization, i.e. $\sim$ 1.0 Ga. McGregor Bay type dikes may well exist within the Grenville Province but we have discovered no criteria to distinguish a remagnetized Sudbury dike from a McGregor Bay type dike.

(6) The Sudbury diabases, in general, do not take on the aspect of metamorphic rocks until a position some 12–14 km within the Grenville Province is reached (sites 74–77 of Fig. 3*b*). Here

TABLE 4. Summary of directional data and derived paleopoles

Rock unit	Direction of cleaned remanence (D, I, N, k)	Pole (N, K, A_{95})
Sudbury dikes (this study)	264.8, +01.7, 38, 27.5	166.9° W, 2.9° S (38, 51, 3.3°)
Sudbury dikes* (Larochelle 1967)	268.2, −00.7, 14, 74.6	170.3° W, 1.5° S (14, 202, 2.8°)
This study and Larochelle (1967) combined		167.8° W, 2.5° S (52, 62, 2.5°)
Sudbury dikes near or within the Grenville Province (this study)	111.2, +26.5, 23, 20.3	16.6° W, 3.7° S (23, 24, 6.3°)
'Lower Paleozoic' dikes near and within the Grenville Province	125.5, +59.1, 3, 64.2	42.3° W, 10.0° N (3, 45, 18.6°)

NOTES: Remanence directional data: D and I are mean declination and inclination calculated from N sites; k is the precision parameter of the mean direction.
Pole data: K and A_{95} are the precision parameter and the radius of the circle of 95% confidence about the mean pole calculated by giving unit weight to N site poles.
*The directional and pole data of this entry differ slightly from that in the original source which was based on giving cores, rather than sites, unit weight.

the stable magnetization, ESE-down, is possibly a genuine high temperature TRM or more probably a combination TRM–CRM (chemical remanent magnetization) produced during the time of 'exsolution' of the Fe–Ti oxides from pyroxene and plagioclase in these rocks.

(7) The K–Ar analyses and the petrochemical study were most fruitful in demonstrating that the ESE-down magnetizations are derived from two populations of magnetization ages. Although we located only three sites from the early Paleozoic population of dikes in our area of study, these tholeiitic diabases may be more abundant within the Grenville. In a study of diabase dikes within the Grenville Province, Murthy (1971) divided his paleomagnetic sites into two groups. Group A sites have a mean remanence direction of 119°, +45° but with considerable streaking in an inclination sense. It would appear probable that Group A includes both early Paleozoic dike sites and those which are remagnetized Sudbury dikes and (or) dikes of 1.0 Ga age equivalent to the McGregor Bay dikes of this study.

Throughout the text and in Table 4 we have referred to the tholeiitic diabases as those with early Paleozoic apparent ages. The apparent ages fall in the Ordovician–Silurian time span yet the mean paleomagnetic pole, (42° W, 10° N) based on 3 sites only, is quite unlike the more reliable Ordovician–Silurian poles for North America as recorded in Irving *et al.* (1976). The pole is similar to some published Cambrian poles but the Cambrian data as a whole are very scattered. Two obvious interpretive possibilities exist: (1) The 'Early Paleozoic dikes' are Cambrian and carry a Cambrian magnetization but the K–Ar ages have been lowered by argon loss. (2) The K–Ar ages reflect the geological age accurately and the magnetization is Ordovician–Silurian but that the Ordovician–Silurian paleofield was more complicated than the available data suggest. We prefer not to state a preference for either of these alternatives with the information at hand.

Acknowledgements

R. L. Wilson provided laboratory facilities at the University of Liverpool where many of the paleomagnetic measurements were made during the early stages of this project. V. Rinco and D. Lacina ably performed many of the paleomagnetic measurements at the University of Western Ontario. Advice concerning the location of critical outcrops in the field was sought from and provided by: K. D. Card, S. B. Lumbers, and M. J. Frarey. B. A. Merz benefited from the guidance given by R. A. Coish and H. W. Hunter in the use of the X-ray fluorescence equipment. C. Herzberg, C. M. Carmichael, and M. J. Frarey, in reviewing a draft manuscript, corrected errors and improved the clarity. We gratefully acknowledge financial support from the National Research Council of Canada through grants A 3544 and A 8163 to Palmer and Hayatsu respectively.

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